ModelIt!

Student Activity Pack

Where Does Rain Come From?

The Amazing Water Cycle

Kindergarten | NGSS K-ESS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Circle where rain comes from: (picture of clouds / picture of a faucet / picture of the ground)

2 Draw a cloud making rain.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sun's Heat

Evaporation Speed

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Cool Day

Sun's Heat set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Hot Day

Sun's Heat set to high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Where does rain come from and where does it go after it falls? Use evidence from your simulation to support your answers.

8 STEM Challenge: Make a Mini Water Cycle

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YOUR ENGINEERING MISSION

Build a mini water cycle in a zip-lock bag to watch water evaporate, form clouds, and rain back down!

THE SCENARIO

Scientists need your help to understand the water cycle! Your team will build a water cycle in a bag so everyone can SEE how water travels from the ground to the sky and back again. Tape it to a sunny window and watch the magic happen!

Engineering Design Thinking

1. HOW MUCH WATER WILL YOU PUT IN THE BAG?
2. WHERE WILL YOU TAPE THE BAG SO THE SUN CAN HEAT IT?
3. HOW WILL YOU DRAW THE WATER CYCLE ON YOUR BAG TO LABEL EACH PART?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists study where water goes — in rivers, underground, in clouds, and in the ocean. They help make sure everyone has clean water to drink! They earn $55,000-$95,000/year.